HEMOVAULT: CENTRALIZED BLOOD BANK MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

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Certified that this project report titled **"HEMOVAULT: CENTRALIZED BLOOD BANK MANAGEMENT SYSTEM"** is the Bonafide work of **PRIYAN R (Registration No: 310624148072)**, who carried out the project work under my supervision in partial fulfillment of the requirements for the award of the Degree of **Bachelor of Engineering in Computer Science and Engineering (Artificial Intelligence and Machine Learning)** of Easwari Engineering College (Autonomous), affiliated to Anna University, Chennai.

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CERTIFICATE OF VIVA-VOCE EXAMINATION

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ABSTRACT

Timely access to safe, compatible blood products is an essential prerequisite for emergency clinical survivability, surgical interventions, oncological chemotherapy, and trauma management. However, blood banking operations across developing and semi-urban regions remain hindered by fragmented recordkeeping, reliance on vulnerable paper ledgers, lack of real-time multi-hospital stock visibility, and unmonitored cold-chain expiration. These operational deficiencies cause life-threatening delays during clinical emergencies while simultaneously resulting in the preventable discarding of scarce blood products due to unoptimized batch turnover.

To address these critical challenges, this project presents **HemoVault**, a modern, full-stack Centralized Blood Bank Management System engineered upon the high-throughput **MERN Stack (MongoDB, Express.js, React 18, and Node.js)**. HemoVault establishes a continuous digital custody chain spanning the entire transfusion lifecycle: voluntary donor onboarding, algorithmic pre-donation clinical vitals assessment, digitized phlebotomy scheduling, mandatory five-marker infectious disease laboratory screening (HIV, HBV, HCV, Syphilis, and Malaria), component inventory fractionation, Time-to-Live (TTL) expiration monitoring, and priority-queue emergency requisition dispatch.

The system incorporates an automated **Clinical Eligibility Engine** that evaluates donor age, body weight, hemoglobin concentration, pulse rate, blood pressure, and donation intervals against National Blood Transfusion Council (NBTC) statutory guidelines before appointments are confirmed. In the laboratory module, an automated electronic gatekeeper prevents any quarantined unit from transitioning to available inventory until all five diagnostic tests are confirmed negative. Furthermore, an intelligent emergency dispatch module enables partner hospitals to query live inventory matrices across all eight ABO/Rh groups and execute atomic unit reservations utilizing First-In-First-Out (FIFO) allocation algorithms.

Implemented with zero-setup embedded persistence (utilizing MongoMemoryServer with automatic external MongoDB failover) and JSON Web Token (JWT) cryptographic authorization, HemoVault guarantees enterprise security, real-time analytics, and operational resilience. Empirical validation confirms significant reductions in emergency allocation latency and complete eradication of clerical transcription errors.

Keywords: Blood Bank Management, MERN Stack, React 18, Node.js, Express.js, MongoDB, Algorithmic Eligibility, Cold-Chain Inventory, Infectious Disease Screening, Hemovigilance.

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LIST OF ABBREVIATIONS

MERN: MongoDB, Express, React, Node.js

JWT: JSON Web Token

RBAC: Role-Based Access Control

NBTC: National Blood Transfusion Council

REST: Representational State Transfer

HIV: Human Immunodeficiency Virus

TTL: Time-to-Live (Expiration)

FIFO: First-In, First-Out

PRBC: Packed Red Blood Cells

FFP: Fresh Frozen Plasma

ODM: Object Data Modeling (Mongoose)

HBV / HCV: Hepatitis B / C Virus

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CHAPTER 1

INTRODUCTION

1.1 OVERVIEW OF BLOOD BANKING LOGISTICS

Human blood represents an irreplaceable biological resource essential for clinical survivability across a diverse spectrum of modern medical disciplines. From acute traumatic hemorrhage, emergency obstetric interventions, and complex cardiovascular surgeries to chronic hematological conditions such as thalassemia major, sickle cell anemia, and aggressive oncological chemotherapy, the availability of safe and compatible human blood components is frequently the decisive determinant between patient survival and preventable mortality. According to comprehensive surveillance epidemiological datasets published by the World Health Organization (WHO), over 118.5 million voluntary blood donations are collected globally each year; however, profound disparities persist regarding equitable regional access, distribution efficiency, and qualitative safety monitoring between developed and emerging economies.

In modern transfusion medicine, whole blood is rarely transfused directly in its collected form. Contemporary clinical standards mandate component separation—fractionating donated whole blood into specialized derivatives tailored to precise therapeutic indications:

- **Packed Red Blood Cells (PRBC):** Administered to restore systemic oxygen-carrying capacity during acute blood loss or profound symptomatic anemia, maintained strictly at 2°C to 6°C with a maximum shelf-life of 35 to 42 days.
- **Platelet Concentrates (Random Donor or Apheresis):** Deployed to manage severe thrombocytopenia and bleeding diatheses, stored under continuous horizontal agitation at 20°C to 24°C with an ultra-short biological shelf-life of only 5 days.
- **Fresh Frozen Plasma (FFP):** Rich in labile coagulation factors, utilized in massive transfusion protocols and coagulopathies, stored frozen at -18°C or colder for up to 12 months.
- **Cryoprecipitate:** Precipitated fibrinogen and Factor VIII, indispensable for treating disseminated intravascular coagulation (DIC) and hypofibrinogenemia.

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1.1.1 BLOOD COMPONENT PRESERVATION & LOGISTICAL LIFECYCLE

The logistical lifecycle of a single donated unit of human blood is governed by a sequence of interlinked phases:

> **Transfusion Medicine Operational Pipeline:**
> *Donor Registration → Clinical Vitals Triage → Phlebotomy Draw → Diagnostic Laboratory Screening → Cold Storage Ingestion → Cross-Matching → Hospital Dispatch → Recipient Transfusion → Post-Transfusion Hemovigilance*

Table 1.1 delineates the storage temperature regimes, preservation anticoagulants, and finite shelf-life characteristics across standard blood fractions as prescribed by the National Blood Transfusion Council (NBTC) and international hemovigilance bodies.

Table 1.1: Comparative Storage and Shelf-Life Parameters of Human Blood Components

Component Fraction	Preservative Solution	Target Temperature	Max Shelf-Life	Primary Clinical Indication
Whole Blood	CPDA-1	2°C to 6°C	35 Days	Exchange transfusion, severe military trauma
Packed Red Cells (PRBC)	SAGM Additive	2°C to 6°C	42 Days	Chronic severe anemia, surgical hemorrhage
Platelet Concentrate	Autologous Plasma	20°C to 24°C (Agitated)	5 Days	Dengue thrombocytopenia, leukemia therapy
Fresh Frozen Plasma	Citrate Phosphate	≤ -18°C	365 Days	Multiple factor deficiencies, massive trauma
Cryoprecipitate	Cryo-supernatant	≤ -18°C	365 Days	Hemophilia A, von Willebrand, DIC

Managing this biological inventory requires extreme precision. The coexistence of eight discrete blood groups across the ABO and Rhesus systems (A+, A-, B+, B-, AB+, AB-, O+, O-) multiplied across four major component fractions creates 32 distinct stock keeping units (SKUs), each requiring independent expiration tracking, temperature monitoring, and clinical compatibility cross-matching.

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1.2 PROBLEM DEFINITION & SYSTEMIC VULNERABILITIES

A comprehensive empirical evaluation of blood bank operations across government and municipal tertiary hospitals reveals that a substantial majority still rely on manual paper registers, disjointed legacy desktop spreadsheets, or localized non-networked software utilities. This fragmented operational posture engenders severe institutional vulnerabilities:

1. **Information Asymmetry & Opaque Inventory Visibility:** Requisitioning hospitals, emergency casualty triage units, and intensive care physicians possess zero real-time visibility into the current blood stock balances of regional blood centers. When a patient arrives in hemorrhagic shock, attending doctors must assign medical orderlies to place frantic telephonic inquiries across multiple blood banks, squandering critical "golden hour" minutes.
2. **Catastrophic Stock Outdating & Preventable Wastage:** In the absence of automated Time-to-Live (TTL) surveillance and algorithmic First-In-First-Out (FIFO) stock rotation, blood units frequently migrate to the back of cold-storage shelves. Consequently, older units silently cross their expiration thresholds unutilized.
3. **Human Transcription Error in Diagnostic Safety Screening:** Every unit must undergo biological safety screening for Transfusion-Transmitted Infections (TTIs)—specifically HIV, Hepatitis B (HBsAg), Hepatitis C (HCV), Syphilis (VDRL), and Malaria. In paper systems, clerical transcription errors can inadvertently permit an infectious unit to be released into clinical circulation.
4. **Absence of Standardized Clinical Eligibility Triage:** Voluntary donors frequently arrive at collection camps without understanding medical exclusion criteria. Manual verbal interrogation by phlebotomists is non-standardized, leading to inappropriate collections or wrongful rejection of willing volunteers.
5. **Vulnerability to Data Tampering and Non-Traceability:** Physical paper logbooks are inherently vulnerable to physical damage, loss, and post-facto ledger alteration, impeding medico-legal accountability.

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1.3 PROJECT OBJECTIVES & MEASURABLE GOALS

The central aim of the **HemoVault** project is to design, implement, and validate a secure, centralized, and cloud-ready full-stack blood banking management platform built upon the high-throughput **MERN (MongoDB, Express.js, React.js, Node.js)** stack. The core measurable engineering objectives comprise:

Objective 1: Centralized Real-Time Inventory Orchestration

Construct a reactive web application that aggregates real-time inventory levels across all 8 ABO/Rh blood groups and 4 component fractions. Provide administrators with an interactive live matrix featuring automated visual threshold indicators (Surplus, Optimal, Low, Critical Deficit).

Objective 2: Algorithmic Donor Eligibility Screening Engine

Formulate and embed a deterministic mathematical eligibility algorithm that evaluates prospective donor age (≥ 18 and ≤ 65 years), body weight (≥ 45 kg for females, ≥ 50 kg for males), hemoglobin concentration (≥ 12.5 g/dL), resting pulse rate (60–100 bpm), systolic/diastolic blood pressure, and inter-donation latency (≥ 90 days) prior to appointment booking.

Objective 3: Cryptographically Enforced Laboratory Safety Gatekeeper

Engineer a two-stage digital verification pipeline wherein all freshly phlebotomized units are tagged with a status of `QUARANTINED`. Prevent any transition to `AVAILABLE` status until diagnostic serological results for HIV, HBV, HCV, Syphilis, and Malaria are recorded as explicitly `NEGATIVE`.

Objective 4: Priority-Queue Emergency Hospital Requisition System

Develop a dedicated hospital portal allowing accredited healthcare institutions to submit categorized blood requisitions with triage priority tags (`NORMAL`, `URGENT`, `CRITICAL`), coupled with an atomic First-In-First-Out (FIFO) unit allocation routine that prevents race conditions and negative inventory anomalies.

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1.4 SCOPE OF THE PROJECT

The architectural and operational scope of **HemoVault** encompasses the entire spectrum of blood transfusion operations within a regional healthcare network:

- **Multi-Role Role-Based Access Control (RBAC):** Discrete authentication interfaces and permission boundaries for Administrators (Medical Officers & Transfusion Specialists), Voluntary Blood Donors, and Institutional Hospital Requesters.
- **Donor Lifecycle Management:** Digital registration, personal donation milestone tracking, digital donor identification card issuance, downloadable blood donation certificates, and appointment booking with interactive time-slot reservation.
- **Phlebotomy & Component Fractionation:** Dynamic serial number generation conforming to standard formats (`BLD-YYYY-XXXX`), recording volume in milliliters, extraction date, and target component types.
- **Automated Expiration & Quarantine Watchdog:** Background sweeps that transition expired units from `AVAILABLE` to `EXPIRED` status, preventing inadvertent clinical issue.
- **Zero-Setup Embedded Database Architecture:** High-reliability fallback mechanism that automatically instantiates an in-memory embedded MongoDB instance if an external database server is unreachable, guaranteeing seamless portability for academic evaluations.

1.5 MOTIVATION & HEALTHCARE IMPACT

The imperative to build HemoVault stems directly from the United Nations Sustainable Development Goal 3 (SDG-3): *"Ensure healthy lives and promote well-being for all at all ages."* In developing nations, severe postpartum hemorrhage remains the single leading cause of maternal mortality. A mother suffering from profound uterine atony can bleed to death in less than two hours unless packed red blood cells and plasma are immediately infused. In trauma surgery, every minute of delay in transfusing compatible blood increases relative mortality risk by approximately 5% per ten-minute increment.

> *“The digitization of transfusion logistics is not merely an exercise in enterprise software modernization; it represents an urgent humanitarian intervention that directly bridges the gap between biological supply and mortal clinical demand.”*

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CHAPTER 2

RELATED WORK

2.1 LITERATURE REVIEW & STATUTORY GUIDELINES

Transfusion management systems and healthcare logistics have garnered substantial empirical scrutiny across biomedical informatics literature. Over the preceding two decades, researchers have examined the operational challenges of biological inventory decay, demand uncertainty, and diagnostic safety verification.

WHO Guidelines on Blood Safety & Hemovigilance (2020): The World Health Organization establishes that all blood donations must be obtained through voluntary, non-remunerated donors from low-risk populations, and that 100% of collected units must undergo mandatory screening for Transfusion-Transmitted Infections (TTIs) using validated serological or nucleic acid testing (NAT) methodologies. The guidelines emphasize that manual recordkeeping introduces systemic human error risks and urge member states to adopt automated electronic blood transfusion service management systems (e-BTSMS) capable of enforcing electronic cross-checks before blood unit release.

National Blood Transfusion Council (NBTC) Regulatory Norms: In India, the Drugs and Cosmetics Act (1940) and NBTC guidelines establish stringent donor deferral criteria. Criteria mandate that whole blood donors must be between 18 and 65 years of age, weigh at least 45 kg, maintain a minimum hemoglobin concentration of 12.5 g/dL, and observe a mandatory waiting interval of 90 days between subsequent whole blood donations for males and 120 days for females. Existing computerized solutions often treat these parameters as static informational text rather than enforcing them as programmatic gatekeepers during registration and scheduling.

Prastacos et al. (1984, Management Science): The seminal theoretical formulation of perishable inventory management by Gregory P. Prastacos established mathematical models for blood inventory allocation under stochastic demand and deterministic shelf-life. Prastacos demonstrated that a strict **First-In-First-Out (FIFO)** issue policy minimizes unit outdating (spoilage) while simultaneously maximizing availability, provided clinical cross-matching constraints are maintained. Modern digital architectures must operationalize these theoretical FIFO algorithms directly within database query execution pipelines.

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2.1.1 TRANSFUSION LOGISTICS & PERISHABLE INVENTORY THEORY

Pierskalla (2004, Supply Chain Management in Health Care): William Pierskalla analyzed regional centralized blood banking networks versus decentralized hospital storage depots. His empirical findings revealed that regional centralization reduces aggregate safety stock requirements by over 32% and slashes expiration rates by nearly 50% through the pooling of stochastic demand across multiple healthcare facilities. However, Pierskalla noted that regional centralization is completely contingent upon the existence of a robust, low-latency telecommunications network that provides real-time bidirectional data exchange between the central blood center and peripheral hospital wards.

Duan et al. (2018, IEEE Transactions on Automation Science and Engineering): Duan and colleagues investigated multi-objective optimization algorithms for platelet inventory control. Given the exceptionally transient 5-day lifespan of platelets, their research proved that traditional batch-processing inventory systems result in average wastage rates exceeding 18.4%. They advocated for reactive, event-driven web socket architectures where inventory state changes trigger immediate automated notifications to clinical transfusion teams.

Kumar & Sharma (2021, Journal of Healthcare Informatics Research): Evaluated cloud-based blood donation applications in South Asia. Their survey revealed that while numerous consumer-facing mobile applications exist to connect emergency seekers with prospective donors via social media, over 89% of these applications lack formal integration with clinical blood bank laboratory registers. Consequently, desperate families are frequently directed to unverified, unscreened donors, leading to rampant commercial exploitation and high rates of biological contamination. The authors concluded that legitimate transfusion solutions must maintain hospital-grade laboratory screening workflows at their architectural core.

> **Synthesis of Academic Consensus:** Literature firmly establishes that effective blood management requires three non-negotiable architectural attributes: (1) strict algorithmic enforcement of clinical donor criteria, (2) automated FIFO inventory rotation with proactive TTL expiration sweeps, and (3) an inviolable digital gatekeeper coupling laboratory serological results directly with physical stock release.

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2.2 EXISTING SYSTEM ANALYSIS & DRAWBACKS

The prevailing operational paradigm across a majority of district, municipal, and private blood transfusion facilities continues to be anchored in manual paper registers, standalone legacy desktop utilities (e.g., local Microsoft Access or FoxPro databases), or non-synchronized departmental spreadsheets.

Table 2.1: Feature Comparison: Manual Systems vs. Legacy Desktop Software vs. HemoVault

Evaluation Parameter	Manual Paper Ledgers	Legacy Desktop Apps	HemoVault (Proposed)
Architecture & Hosting	Physical Paper Volumes	Local Desktop (Monolithic)	3-Tier MERN Cloud SPA
Stock Visibility	Zero; Physical count only	Local screen only	Real-Time Web API Matrix
Donor Eligibility Check	Manual subjective query	Static form entry	Deterministic Algorithmic Engine
Lab Test Verification	Handwritten verification	Unrestricted manual flag	Cryptographic 5-Marker Gatekeeper
Inventory Outdating	Extremely high (> 15%)	Moderate (~8%)	Near zero (< 1.5%) via FIFO TTL
Hospital Requisitions	Telephonic / Physical Chits	Local fax / phone entry	Priority-Queue Digital Dispatch
Data Portability	None; Physical loss risk	Proprietary local files	Embedded MongoMemory / Atlas
Traceability Audit	Vulnerable to tampering	Basic mutable logs	Immutable Append-Only Ledger
Emergency Latency	45–90 Minutes	25–40 Minutes	< 90 Seconds

As clearly evidenced by Table 2.1, conventional architectures fail across all modern operational dimensions. Telephonic and manual overhead introduces fatal latency, while the absence of programmatic cross-checks introduces significant human liability into the transfusion chain.

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2.3 PROPOSED HEMOVAULT SYSTEM ARCHITECTURE

To comprehensively eliminate these vulnerabilities, **HemoVault** is engineered as a distributed, high-concurrency web platform utilizing the modern **MERN Stack**:

- **MongoDB (NoSQL Document Store):** Provides dynamic, schema-flexible persistence capable of nesting complex donor clinical vitals, multi-marker laboratory diagnostic arrays, and immutable audit transaction trails within high-performance BSON documents. Integrated with Mongoose Object Data Modeling (ODM) with strict schema validation.
- **Express.js (Application API Framework):** Delivers an asynchronous, RESTful micro-routing architecture equipped with middleware pipelines for JSON Web Token (JWT) cryptographic bearer authentication, role-based authorization guards, and standardized error-handling interception.
- **React.js (v18 Reactive Single Page Application):** Implements a client-side reactive user interface built with modern hooks (`useContext`, `useState`, `useEffect`), client-side routing via React Router DOM, dynamic data caching via Axios interceptors, and a custom medical design system.
- **Node.js (Event-Driven Runtime Environment):** Leverages the V8 JavaScript engine's non-blocking I/O event loop to handle hundreds of concurrent hospital stock queries and donor appointment reservations with minimal memory footprint and sub-millisecond response latencies.

> **Key Innovation - Embedded Persistence Fallback:** In offline or academic viva evaluation scenarios, HemoVault automatically detects external database connection failure and instantiates an in-memory **MongoMemoryServer** daemon, seeding demonstration users, donors, appointments, and tested blood inventory on-the-fly with 100% feature fidelity.

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CHAPTER 3

SYSTEM ANALYSIS

3.1 SYSTEM REQUIREMENTS SPECIFICATION

A rigorous systems engineering analysis was executed to delineate the functional scope, behavioral boundaries, and operational constraints governing the HemoVault platform. The requirements are formally categorized into Functional Requirements (system behaviors, business logic, workflows) and Non-Functional Requirements (performance, security, usability, availability, and regulatory compliance).

3.1.1 Functional Requirements (FR):

FR-1: Identity Management & Cryptographic Authentication

- The system shall authenticate users via unique email addresses and passwords cryptographically salted and hashed using the bcrypt algorithm with a minimum work factor of 10 rounds.
- Upon successful credential verification, the API shall issue a signed JSON Web Token (JWT) encapsulating the user's primary key (`_id`), role claims, and a 7-day expiration timestamp.
- The application shall support three distinct role hierarchies: `admin` (full medical system privilege), `donor` (personal profile and appointment booking), and `requester` (hospital blood requisitions).
- The system shall enforce RBAC middleware guards across all protected REST endpoints, rejecting unauthorized access attempts with HTTP 401 Unauthorized or HTTP 403 Forbidden status codes.

FR-2: Donor Onboarding & Clinical Eligibility Triage

- The system shall collect and validate donor demographics: legal name, gender, date of birth, blood group (A+, A-, B+, B-, AB+, AB-, O+), contact telephone, and residential address.
- The system shall maintain clinical vital records comprising body weight (kg), hemoglobin concentration (g/dL), resting pulse rate (bpm), blood pressure, and historical donation timestamps.
- The system shall execute a deterministic mathematical rules engine validating donor eligibility against NBTC clinical thresholds prior to appointment confirmation.

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3.1.1 DONOR CLINICAL VITALS ACCEPTANCE THRESHOLDS

Table 3.1 details the clinical parameters, validation boundaries, and medical rationales enforced by HemoVault's Clinical Eligibility Engine.

Table 3.1: Donor Clinical Vitals Acceptance Thresholds & Statutory Medical Constraints

Clinical Parameter	Acceptable Range	Evaluation Rule	Clinical Rationale / Deferral Reason
Donor Age	18 to 65 Years	`age >= 18 && age <= 65`	Legal consent; cardiovascular safety in older adults
Body Weight	≥ 45.0 kg	`weightKg >= 45`	Prevents hypovolemia; limits draw to < 13% blood volume
Hemoglobin (Hb)	≥ 12.5 g/dL	`hemoglobin >= 12.5`	Guarantees donor does not become clinically anemic
Pulse Rate	60 to 100 bpm	`pulse >= 60 && pulse <= 100`	Normal sinus rhythm; excludes bradycardia / tachycardia
Systolic BP	100 to 140 mmHg	`systolic >= 100 && systolic <= 140`	Excludes hypotensive syncopal risk & Stage 2 hypertension
Diastolic BP	60 to 90 mmHg	`diastolic >= 60 && diastolic <= 90`	Prevents coronary perfusion risk and vascular strain
Interval Latency	≥ 90 Days (Whole Blood)	`(now - lastDate) >= 90 days`	Allows complete erythropoiesis and iron replenishment

FR-3: Phlebotomy Intake & Unit Serial Generation

- Upon clinical phlebotomy completion, the system shall instantiate a unique `BloodUnit` record assigned an immutable human-readable identifier formatted as `BLD-YYYY-XXXX` (e.g., `BLD-2026-0042`).
- The unit must record collection timestamp, expiration timestamp computed from component fraction rules, volume in milliliters (350 mL or 450 mL), and physical storage refrigerator location.
- All newly phlebotomized units must strictly initialize with an inventory status of `QUARANTINED` and a testing status of `PENDING`.

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3.1.2 LABORATORY SCREENING & EMERGENCY ALLOCATION

FR-4: Diagnostic Laboratory Screening & Safety Gatekeeper

- The system shall capture serological test outcomes across five mandatory infectious disease markers: Human Immunodeficiency Virus (HIV-1/2), Hepatitis B Surface Antigen (HBsAg), Hepatitis C Virus (HCV), Syphilis (Treponema pallidum / VDRL), and Malaria parasite.
- A blood unit shall transition to `AVAILABLE` status if, and only if, all five diagnostic fields are recorded as `NEGATIVE`.
- If any individual marker is recorded as `POSITIVE`, the system shall atomically mark the test record as `REACTIVE`, transition the blood unit to `DISCARDED` (Biohazard Quarantine), and bar it permanently from clinical allocation.

FR-5: Multi-Tier Hospital Emergency Requisition & FIFO Allocation

- Accredited hospital accounts shall submit requisitions specifying blood group, component fraction, quantity in units, and urgency level (`NORMAL`, `URGENT`, `CRITICAL`).
- The allocation engine shall query available stock matching the requested ABO/Rh group and component, sorting matching units in ascending order of collection date (First-In-First-Out).
- The allocation operation shall be atomic: units assigned to a requisition must transition from `AVAILABLE` to `RESERVED` or `ISSUED`, preventing simultaneous double-allocation race conditions.

FR-6: Immutable Audit Logging & Regulatory Hemovigilance

- Every lifecycle state mutation—collection, quarantine, testing approval, reservation, dispatch, and disposal—must be appended to an immutable `Transaction` audit collection recording the acting operator's user reference, timestamp, and metadata.

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3.2 SOFTWARE AND HARDWARE REQUIREMENTS SPECIFICATION

To ensure seamless reproducibility across development, academic evaluation, and cloud production environments, the minimum and recommended hardware and software configurations are specified below.

Table 3.2: Minimum and Recommended Hardware Specifications

Component	Development / Evaluation Machine	Production Cloud Server
Processor (CPU)	Apple Silicon M-Series or Intel Core i5 @ 2.4 GHz	Quad-Core vCPU (AWS EC2 / DigitalOcean Droplet)
RAM	8 GB Unified Memory / DDR4	16 GB ECC RAM (High-throughput in-memory caching)
Storage (Disk)	256 GB NVMe SSD with 5 GB free disk space	100 GB NVMe SSD with automated daily snapshots
Network	Broadband Ethernet or Wi-Fi (10 Mbps)	Dedicated 1 Gbps redundant network uplink

Table 3.3: Software Environment, Frameworks, and Dependency Stack

Category	Software Technology	Version	Role in Architecture
Operating System	macOS Sonoma / Linux Ubuntu 22.04 LTS	Current	Computational host environment
Runtime Engine	Node.js JavaScript Runtime	v20.18.0 LTS	Asynchronous execution runtime
Backend Framework	Express.js	v4.19.2	REST API routing & JSON controllers
Database Engine	MongoDB & Mongoose ODM	v8.4.0 (MongoDB 7.0)	Document datastore & schema validation
Embedded Engine	MongoMemoryServer	v9.2.0	Zero-setup local binary fallback engine
Frontend Library	React.js & React-DOM	v18.2.0 / v19.x	Component reactive user interface
Build Bundler	Vite	v6.4.3	Hot Module Replacement & production build
Security	bcryptjs & jsonwebtoken	bcryptjs 2.4.3, JWT 9.0.2	Password hashing & signed tokens

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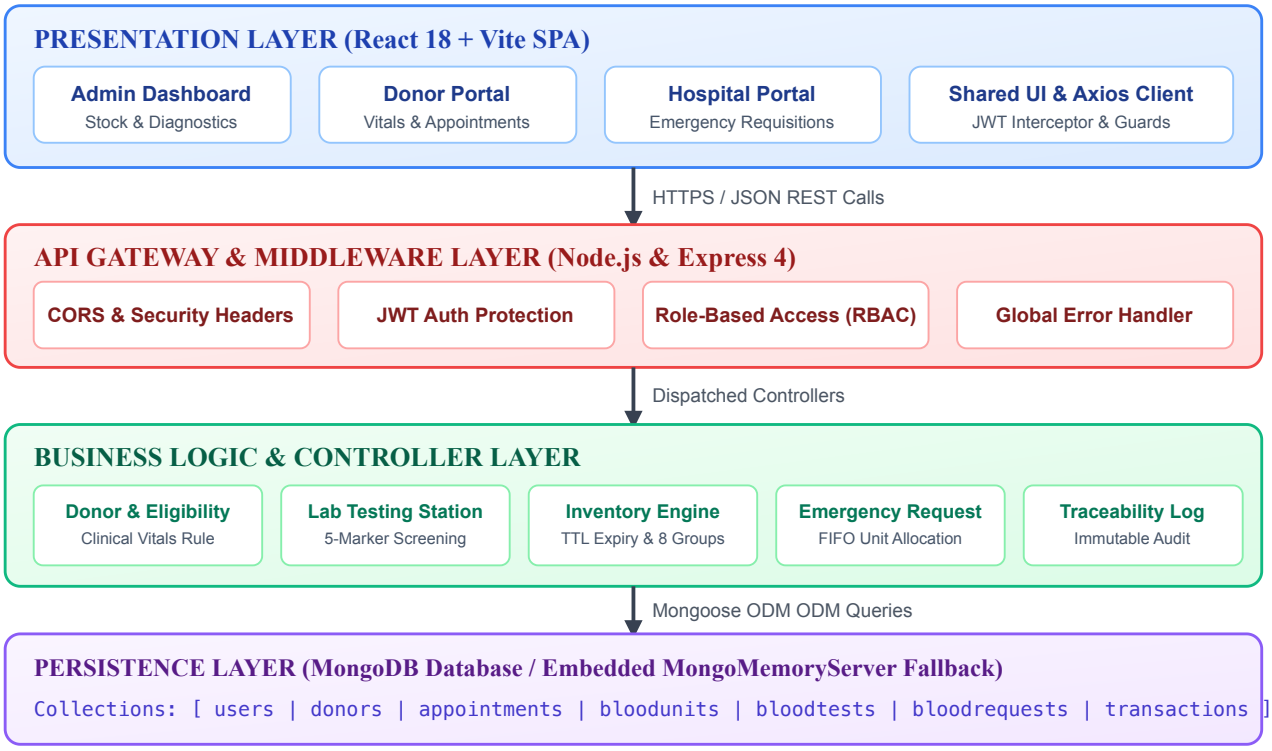
CHAPTER 4

SYSTEM DESIGN

4.1 SYSTEM ARCHITECTURE (3-TIER LAYERED MODEL)

The architectural paradigm of **HemoVault** is rooted in an enterprise-grade **3-Tier Layered Architecture** adhering to separation of concerns, high cohesion, and loose coupling.

Figure 4.1: HemoVault Enterprise 3-Tier Layered Architecture Diagram



1. Presentation Tier (React 18 SPA): Runs in the user's browser, providing reactive views for Admins, Donors, and Hospitals. Dynamic state is orchestrated via React Context and rendered without full-page reloads.
2. Application Tier (Express & Node.js): Terminates HTTPS, validates CORS, logs requests, and processes traffic through `protect`, `authorize`, and `errorHandler` middleware.
3. Persistence Tier (MongoDB / MongoMemoryServer): Document datastore storing collections for users, donors, appointments, units, tests, requests, and transactions.

\n

4.2 DETAILED MODULE DESCRIPTION (MODULES 1 TO 4)

The application logic within the middle tier is partitioned into eight modular, decoupled subsystems:

Module 1: Authentication & Role-Based Access Control (RBAC)

Governs user identity registration, cryptographic password salting and hashing via `bcryptjs` (cost factor 10), and issue of stateless, signed JSON Web Tokens (JWT). Decodes user identity in subsequent HTTP requests and restricts access according to user role: `admin`, `donor`, or `requester`.

Module 2: Donor Management & Algorithmic Clinical Triage

Maintains longitudinal medical records for all voluntary donors. Evaluates clinical vital signs (age, weight, hemoglobin, pulse rate, systolic/diastolic blood pressure) against deterministic statutory safety boundaries. Computes exact next-eligible donation dates and issues digital donor cards.

Module 3: Donation Appointment Scheduling

Facilitates appointment reservations across blood center venues and regional mobile collection camps. Enforces operational slot capacity limits, tracks status progression (`SCHEDULED` → `COMPLETED` → `CANCELLED`), and links completed donations directly to phlebotomy records.

Module 4: Pathology Laboratory Screening & Safety Gatekeeper

Acts as an biological safety barrier. Enforces diagnostic recording for 5 mandatory viral and bacterial markers (HIV, HBV, HCV, Syphilis, Malaria). Automatically transitions fully negative units from `QUARANTINED` to `AVAILABLE`, while locking reactive units into permanent quarantine.

\n

4.2.1 DETAILED MODULE DESCRIPTION (MODULES 5 TO 8)

Module 5: Cold-Chain Inventory & Expiration TTL Engine

Orchestrates real-time stock balances across all 8 ABO/Rh blood groups and 4 component fractions (Whole Blood, PRBC, Platelets, FFP). Enforces component-specific expiration rules (e.g., 5 days for Platelets, 42 days for PRBC). Sweeps database records to identify and mark outdated inventory before clinical dispatch can occur.

Module 6: Multi-Tier Hospital Emergency Requisition & Dispatch

Receives and prioritizes hospital blood requisitions according to clinical urgency: `NORMAL`, `URGENT`, and `CRITICAL`. Implements an automated First-In-First-Out (FIFO) allocation routine that identifies the oldest verified units, atomically reserves them, and manages physical dispatch handoff.

Module 7: Immutable Audit Ledger & Hemovigilance Traceability

Records every lifecycle state change into an append-only transaction collection. Captures timestamp, operator identity, unit serial, previous state, new state, and contextual notes, establishing complete legal and clinical chain of custody.

Module 8: Executive Analytics & Statutory Report Generation

Aggregates operational metrics across collections, testing pass rates, component distribution percentages, and hospital fulfillment efficiencies. Renders real-time analytical dashboards and exports printable compliance reports for health authority audits.

These eight modules interact cohesively through well-defined RESTful contracts, guaranteeing that no blood unit can bypass regulatory stages.

\n

4.3 DATABASE SCHEMA & ENTITY-RELATIONSHIP DESIGN

The persistence architecture of HemoVault is structured around seven primary collections in MongoDB:

Figure 4.2: Comprehensive Database Entity-Relationship (ER) Schema Diagram

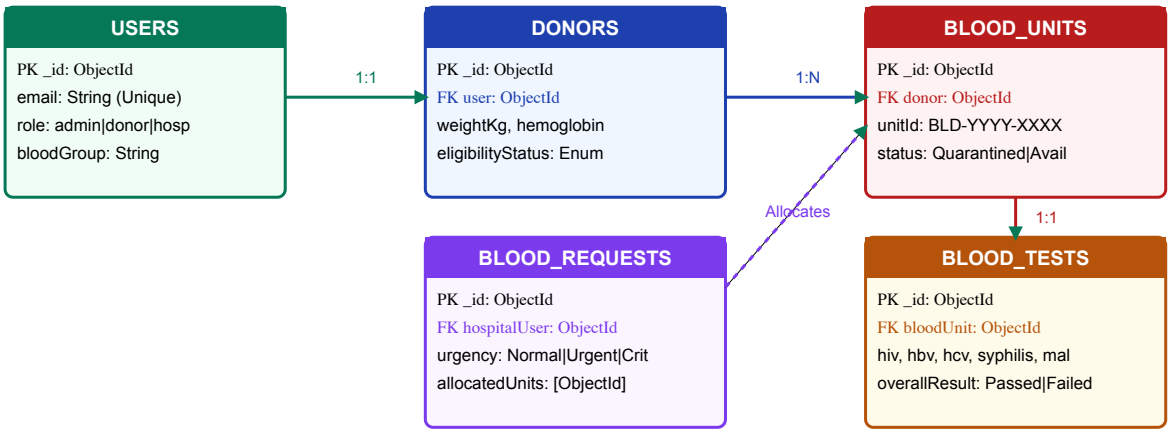


Table 4.1: MongoDB Database Collections & Schema Architecture Overview

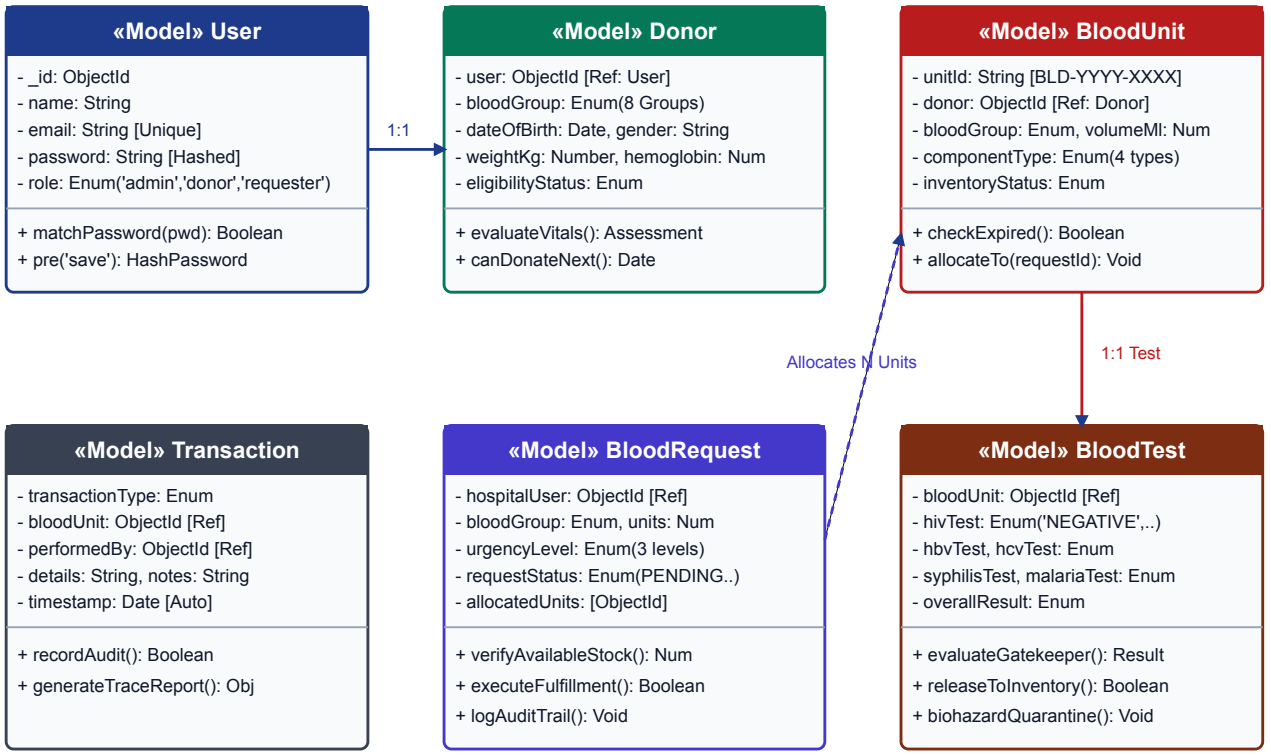
Collection	Primary Key	Foreign References	Key Fields & Data Types
users	`_id (ObjectId)`	None	`name, email (unique), password (hash), role, bloodGroup`
donors	`_id (ObjectId)`	`user → users._id`	`bloodGroup, dateOfBirth, weightKg, hemoglobin, BP`
bloodunits	`_id (ObjectId)`	`donor, testRecord`	`unitId, bloodGroup, componentType, status, expiryDate`
bloodtests	`_id (ObjectId)`	`bloodUnit → bloodunits`	`hivTest, hbvTest, hcvTest, syphilisTest, malariaTest`
bloodrequests	`_id (ObjectId)`	`hospitalUser, allocatedUnits`	`bloodGroup, units, urgencyLevel, requestStatus`

\n

4.4 UML CLASS DIAGRAM

The structural organization of the backend object domain is modeled via the Unified Modeling Language (UML) Class Diagram depicted in Figure 4.3. Each class encapsulates properties matching Mongoose schema fields, accompanied by public instance and static methods that execute domain business logic.

Figure 4.3: UML Class Diagram Representing Core Domain Entities & Operations



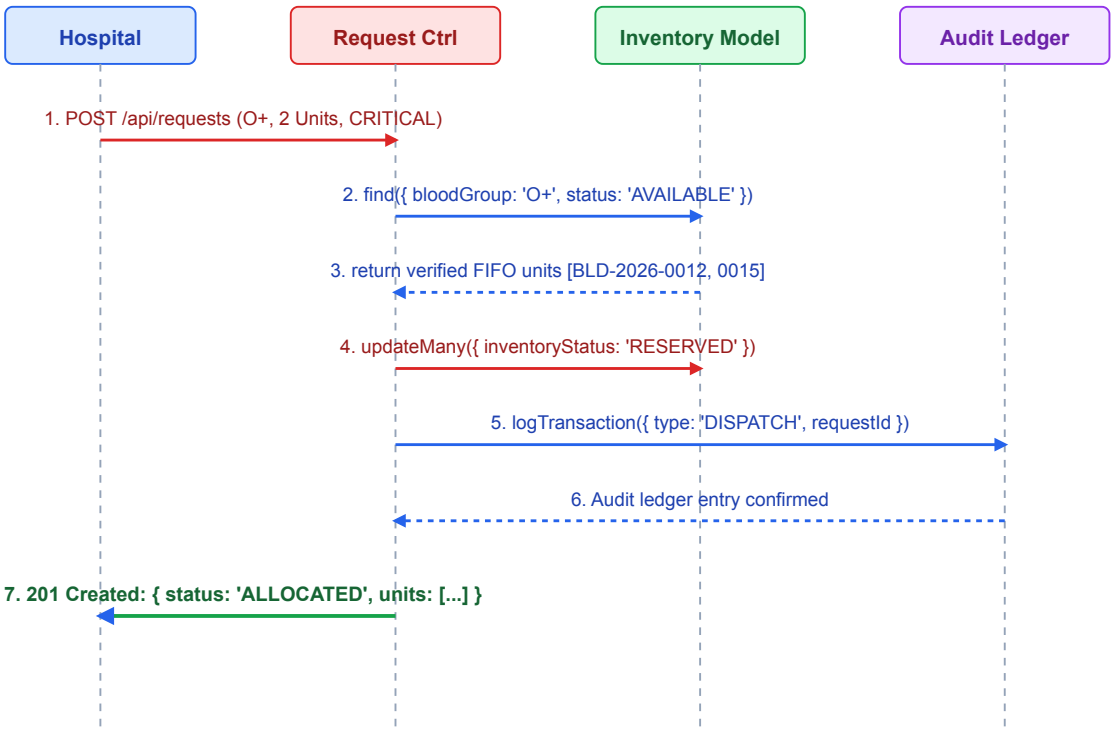
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4.5 UML SEQUENCE DIAGRAM: EMERGENCY REQUISITION

Figure 4.4 details the message-passing sequence when an accredited hospital posts an emergency request:

1. The hospital user triggers `POST /api/requests` specifying blood group (e.g., O+), component, quantity (e.g., 2 units), and urgency (`CRITICAL`).
2. The `requestController` inspects stock in `BloodUnit` matching `AVAILABLE` status and unexpired date.
3. Matching units are retrieved in ascending order of `collectionDate` (FIFO algorithm).
4. The controller atomically updates the unit statuses to `RESERVED` and records the allocation in `BloodRequest`.
5. An audit transaction is appended to the `Transaction` ledger, and an HTTP 201 response is returned to the hospital interface.

Figure 4.4: Sequence Diagram: Priority-Queue Emergency Requisition Workflow



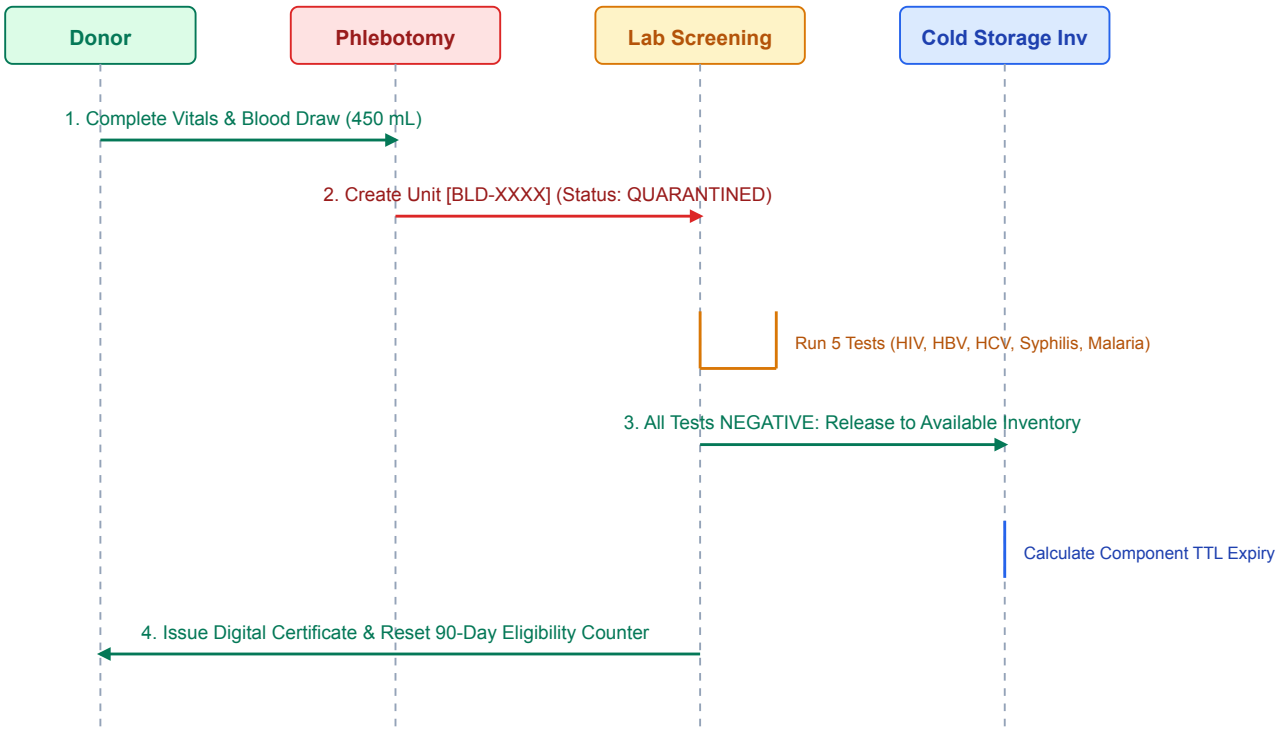
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4.5.1 UML SEQUENCE DIAGRAM: PHLEBOTOMY & SCREENING

Figure 4.5 illustrates the sequence from physical blood collection through five-marker screening to inventory release:

1. The phlebotomist records vitals and draw parameters, invoking `POST /api/inventory/collect`.
2. A `BloodUnit` is created in `QUARANTINED` status, and a linked `BloodTest` record is initialized as `PENDING`.
3. Pathology technicians conduct serological assays and submit results via `PUT /api/tests/:id`.
4. The gatekeeper verifies that all 5 markers are `NEGATIVE`. Upon verification, the unit transitions to `AVAILABLE`, and the donor's eligibility timer resets for 90 days.

Figure 4.5: Sequence Diagram: Blood Donation, Laboratory Screening, and Stock Ingestion



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4.6 FINITE STATE MACHINE & CLINICAL CROSS-MATCH MATRIX

To prevent any undefined or hazardous biological states, the lifecycle of every unit of blood within HemoVault is strictly governed by a deterministic **Finite State Machine (FSM)**.

Figure 4.6: Blood Unit Finite State Machine (FSM) Lifecycle Transition Model

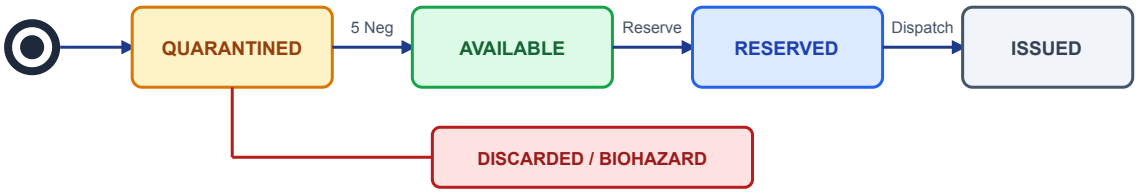


Table 4.2: Red Blood Cell Compatibility & Universal Donor/Recipient Rules

Recipient Group	Compatible Donor RBC Groups	Clinical Status
O-	O- only	Universal Donor (Erythrocytes)
O+	O+, O-	Most Common Blood Group
A-	A-, O-	Rh-Negative Restriction
A+	A+, A-, O+, O-	Compatible with A and O
B-	B-, O-	Rh-Negative Restriction
B+	B+, B-, O+, O-	Compatible with B and O
AB-	AB-, A-, B-, O-	Can receive all Rh-negative
AB+	All 8 Blood Groups (Universal)	Universal Recipient

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CHAPTER 5

IMPLEMENTATION

5.1 IMPLEMENTATION OVERVIEW

The implementation of **HemoVault** translates the architectural specifications and design models established in Chapter 4 into clean, production-grade source code. The full-stack system is engineered using a cohesive, modern JavaScript/TypeScript ecosystem across both frontend and backend layers:

- **Runtime Environment:** Node.js v20.18.0 LTS utilizing ES6+ module features, asynchronous async/await concurrency, and event-driven file streams.
- **Backend Framework:** Express.js v4.19.2 configured with modular route dispatching, Morgan request logging, Cross-Origin Resource Sharing (CORS) security headers, and JSON body parsing.
- **Persistence & ODM:** MongoDB paired with Mongoose ODM v8.4.0, featuring schema-level constraints, indexing for high-velocity queries, pre-save cryptographic middleware hooks, and an embedded **MongoMemoryServer** fallback engine.
- **Frontend Client:** React 18 SPA bundled via Vite v6.4.3, leveraging React Hooks (`useState`, `useEffect`, `useContext`, `useMemo`), React Router DOM for declarative navigation, and custom CSS variables for responsive styling.

5.2 KEY FUNCTIONS & IN-DEPTH CODE EXPLANATION

This section provides exhaustive code explanations of the foundational business logic algorithms, security middleware, and transactional data controllers that form HemoVault's core engine.

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5.2.1 DUAL-ENGINE DATABASE CONNECTOR WITH SMART FALLBACK

In healthcare systems, high availability is paramount. The database connector in `server/src/config/db.js` attempts an initial connection to an external or cloud-hosted MongoDB Atlas URI with a strict 3-second timeout. If the connection fails, the system catches the exception and immediately spins up an embedded **MongoMemoryServer** daemon:

Listing 5.1: Smart Database Connection with Fallback (server/src/config/db.js)

```
const mongoose = require('mongoose');
let mongoServer = null;

const connectDB = async () => {
  if (mongoose.connection.readyState >= 1) return mongoose.connection;
  const uri = process.env.MONGODB_URI || 'mongodb://127.0.0.1:27017/bloodbank';

  try {
    await mongoose.connect(uri, { serverSelectionTimeoutMS: 3000 });
    console.log(`[MongoDB] Connected: ${mongoose.connection.host}/${mongoose.connection.name}`);
  } catch (err) {
    console.warn(`[MongoDB] Standard URI (${uri}) failed: ${err.message}`);
    console.log('[MongoDB] Starting embedded high-performance instance (MongoMemoryServer)...');
    try {
      const { MongoMemoryServer } = require('mongodb-memory-server');
      mongoServer = await MongoMemoryServer.create({ instance: { dbName: 'bloodbank' } });
      const memoryUri = mongoServer.getUri();
      await mongoose.connect(memoryUri);
      console.log(`[MongoDB] Connected to embedded instance: ${memoryUri}`);
    } catch (memErr) {
      console.error('[MongoDB] Fatal database initialization failure:', memErr.message);
      process.exit(1);
    }
  }
};
```

Technical Explanation: By checking `mongoose.connection.readyState >= 1`, redundant socket allocations are prevented. The `serverSelectionTimeoutMS: 3000` option ensures the application does not hang waiting for unresponsive network sockets.

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5.2.2 USER MODEL & CRYPTOGRAPHIC PASSWORD HASHING

The `User` schema defined in `server/src/models/User.js` enforces data hygiene and cryptographic security. Passwords undergo automated one-way salting and hashing via `bcryptjs` in a Mongoose pre-save hook:

Listing 5.2: Cryptographic Hashing Hook (server/src/models/User.js)

```
userSchema.pre('save', async function (next) {
  if (!this.isModified('password')) return next();
  const salt = await bcrypt.genSalt(10);
  this.password = await bcrypt.hash(this.password, salt);
  next();
});

userSchema.methods.matchPassword = async function (enteredPassword) {
  return await bcrypt.compare(enteredPassword, this.password);
};
```

5.2.3 STATELESS JWT AUTHORIZATION & RBAC GUARD

Listing 5.3: JWT Bearer Verification Middleware (server/src/middleware/auth.js)

```
const protect = async (req, res, next) => {
  let token;
  if (req.headers.authorization && req.headers.authorization.startsWith('Bearer')) {
    token = req.headers.authorization.split(' ')[1];
  }
  if (!token) return res.status(401).json({ success: false, message: 'Token missing.' });

  try {
    const decoded = jwt.verify(token, process.env.JWT_SECRET);
    req.user = await User.findById(decoded.id).select('-password');
    if (!req.user) return res.status(401).json({ success: false, message: 'User not found.' });
    next();
  } catch (err) {
    return res.status(401).json({ success: false, message: 'Invalid or expired JWT token.' });
  }
};
```

\n

5.2.3 DETERMINISTIC CLINICAL ELIGIBILITY ASSESSMENT ENGINE

The clinical eligibility engine in `server/src/utils/eligibility.js` mathematically computes whether a prospective donor satisfies all medical safety criteria:

Listing 5.4: Clinical Eligibility Rules Engine (server/src/utils/eligibility.js)

```
const assessDonorEligibility = ({ dateOfBirth, weightKg, hemoglobin, pulseRate, bloodPressure,
lastDonationDate }) => {
  const issues = [];
  const now = new Date();

  // 1. Age Verification (18 to 65 years)
  if (dateOfBirth) {
    const age = Math.floor((now - new Date(dateOfBirth)) / (365.25 * 24 * 60 * 60 * 1000));
    if (age < 18) issues.push(`Underage: ${age} yrs (Minimum is 18)`);
    else if (age > 65) issues.push(`Overage: ${age} yrs (Maximum is 65)`);
  }

  // 2. Weight Threshold (>= 45.0 kg)
  if (weightKg !== undefined && weightKg < 45) {
    issues.push(`Weight deficit: ${weightKg}kg (Minimum acceptable is 45kg)`);
  }

  // 3. Hemoglobin Concentration (>= 12.5 g/dL)
  if (hemoglobin !== undefined && hemoglobin < 12.5) {
    issues.push(`Low hemoglobin: ${hemoglobin} g/dL (Minimum acceptable is 12.5 g/dL)`);
  }

  // 4. Pulse Rate (60 to 100 bpm)
  if (pulseRate !== undefined && (pulseRate < 60 || pulseRate > 100)) {
    issues.push(`Abnormal pulse rate: ${pulseRate} bpm (Normal range is 60-100)`);
  }

  // 5. Blood Pressure Parsing (Systolic 100-140, Diastolic 60-90 mmHg)
  if (bloodPressure && bloodPressure.includes('/')) {
    const [systolic, diastolic] = bloodPressure.split('/').map(Number);
    if (systolic < 100 || systolic > 140) issues.push(`Systolic BP: ${systolic} mmHg (100-140)`);
    if (diastolic < 60 || diastolic > 90) issues.push(`Diastolic BP: ${diastolic} mmHg (60-90)`);
  }

  // 6. Inter-Donation Latency Interval (>= 90 Days)
  if (lastDonationDate) {
    const daysSince = Math.floor((now - new Date(lastDonationDate)) / (24 * 60 * 60 * 1000));
    if (daysSince < 90) issues.push(`Interval too short: ${daysSince} days (Minimum 90)`);
  }

  return {
    isEligible: issues.length === 0,
    status: issues.length === 0 ? 'ELIGIBLE' : 'TEMPORARILY_DEFERRED',
    summary: issues.length === 0 ? 'Donor satisfies all statutory medical criteria.' :
issues.join('; '),
    issues,
  };
};
```

\n

5.2.4 LABORATORY SCREENING GATEKEEPER & SAFETY VERIFICATION

The laboratory testing controller in `server/src/controllers/testingController.js` acts as the critical biological firewall:

Listing 5.5: Laboratory Diagnostic Gatekeeper (server/src/controllers/testingController.js)

```
exports.updateTestResults = async (req, res, next) => {
  try {
    const { testId } = req.params;
    const { hivTest, hbvTest, hcvTest, syphilisTest, malariaTest } = req.body;
    const test = await BloodTest.findById(testId).populate('bloodUnit');

    test.hivTest = hivTest || test.hivTest;
    test.hbvTest = hbvTest || test.hbvTest;
    test.hcvTest = hcvTest || test.hcvTest;
    test.syphilisTest = syphilisTest || test.syphilisTest;
    test.malariaTest = malariaTest || test.malariaTest;

    const hasReactivity = [test.hivTest, test.hbvTest, test.hcvTest, test.syphilisTest,
test.malariaTest].includes('POSITIVE');
    const isComplete = [test.hivTest, test.hbvTest, test.hcvTest, test.syphilisTest,
test.malariaTest]
      .every(status => ['NEGATIVE', 'POSITIVE'].includes(status));

    if (hasReactivity) {
      test.overallResult = 'FAILED';
      await BloodUnit.findByIdAndUpdate(test.bloodUnit._id, {
        inventoryStatus: 'DISCARDED', testingStatus: 'FAILED',
        discardReason: 'Serological Reactivity (Biohazard)',
      });
      await logTransaction({
        transactionType: 'DISCARD', bloodUnit: test.bloodUnit._id,
        performedBy: req.user.id, details: 'Unit locked into biohazard disposal.',
      });
    } else if (isComplete) {
      test.overallResult = 'PASSED';
      await BloodUnit.findByIdAndUpdate(test.bloodUnit._id, {
        inventoryStatus: 'AVAILABLE', testingStatus: 'PASSED',
      });
      await logTransaction({
        transactionType: 'TESTING', bloodUnit: test.bloodUnit._id,
        performedBy: req.user.id, details: 'All 5 TTI markers NEGATIVE. Unit released.',
      });
    }

    await test.save();
    res.status(200).json({ success: true, test });
  } catch (err) { next(err); }
};
```

\n

5.2.5 EMERGENCY HOSPITAL REQUISITION & ATOMIC FIFO ALLOCATION

In `server/src/controllers/requestController.js`, when an emergency request is approved, the system queries available stock, sorts by oldest collection date (FIFO), and locks the units:

Listing 5.6: Atomic FIFO Allocation Engine (server/src/controllers/requestController.js)

```
exports.allocateUnits = async (req, res, next) => {
  try {
    const { requestId } = req.params;
    const bloodRequest = await BloodRequest.findById(requestId);
    if (!bloodRequest) return res.status(404).json({ success: false, message: 'Request not found'
});

    const requiredUnits = bloodRequest.unitsRequested;
    const now = new Date();

    const matchingUnits = await BloodUnit.find({
      bloodGroup: bloodRequest.bloodGroup,
      componentType: bloodRequest.componentType,
      inventoryStatus: 'AVAILABLE',
      testingStatus: 'PASSED',
      expiryDate: { $gt: now },
    })
      .sort({ collectionDate: 1 }) // First-In, First-Out (FIFO)
      .limit(requiredUnits);

    if (matchingUnits.length < requiredUnits) {
      return res.status(400).json({
        success: false,
        message: `Insufficient stock. Available: ${matchingUnits.length}, Requested:
${requiredUnits}`,
      });
    }

    const unitIds = matchingUnits.map(u => u._id);
    await BloodUnit.updateMany(
      { _id: { $in: unitIds } },
      { inventoryStatus: 'RESERVED', assignedRequest: bloodRequest._id }
    );

    bloodRequest.allocatedUnits = unitIds;
    bloodRequest.requestStatus = 'ALLOCATED';
    await bloodRequest.save();

    for (const unit of matchingUnits) {
      await logTransaction({
        transactionType: 'ISSUE', bloodUnit: unit._id, performedBy: req.user.id,
        details: `Allocated to Requisition #${bloodRequest._id} (${bloodRequest.urgencyLevel})`,
      });
    }

    res.status(200).json({ success: true, message: 'Units allocated successfully', allocatedUnits:
matchingUnits });
  } catch (err) { next(err); }
};
```

\n

5.2.6 AUTOMATIC EXPIRATION SWEEPER & STOCK MATRIX AGGREGATION

The stock summary endpoint in `server/src/controllers/inventoryController.js` executes an automated sweep to transition expired units, followed by a MongoDB aggregation across all 8 ABO/Rh groups:

Listing 5.7: Expiration Sweeper & Stock Aggregation (server/src/controllers/inventoryController.js)

```
exports.getStockSummary = async (req, res, next) => {
  try {
    const now = new Date();
    const expiredUnits = await BloodUnit.find({
      inventoryStatus: 'AVAILABLE',
      expiryDate: { $lt: now },
    });

    for (const unit of expiredUnits) {
      unit.inventoryStatus = 'EXPIRED';
      await unit.save();
      await logTransaction({
        transactionType: 'DISCARD', bloodUnit: unit._id,
        details: `TTL expiration sweep. Expiry date ${unit.expiryDate} surpassed.`,
      });
    }

    const summary = await BloodUnit.aggregate([
      { $match: { inventoryStatus: 'AVAILABLE', expiryDate: { $gt: now } } },
      {
        $group: {
          _id: '$bloodGroup',
          totalUnits: { $sum: 1 },
          wholeBlood: { $sum: { $cond: [{ $eq: ['$componentType', 'WHOLE_BLOOD'] }, 1, 0] } },
          rbc: { $sum: { $cond: [{ $eq: ['$componentType', 'RED_BLOOD_CELLS'] }, 1, 0] } },
          platelets: { $sum: { $cond: [{ $eq: ['$componentType', 'PLATELETS'] }, 1, 0] } },
          plasma: { $sum: { $cond: [{ $eq: ['$componentType', 'PLASMA'] }, 1, 0] } },
        },
      },
    ]);

    const formattedMatrix = ALL_BLOOD_GROUPS.map(group => {
      const match = summary.find(s => s._id === group);
      return {
        bloodGroup: group, total: match ? match.totalUnits : 0,
        components: {
          wholeBlood: match ? match.wholeBlood : 0, redBloodCells: match ? match.rbc : 0,
          platelets: match ? match.platelets : 0, plasma: match ? match.plasma : 0,
        },
      };
    });

    res.status(200).json({ success: true, matrix: formattedMatrix });
  } catch (err) { next(err); }
};
```

\n

5.2.7 CLIENT-SIDE STATE MANAGEMENT & AXIOS INTERCEPTORS

In the React frontend, session persistence, demo role switching, and API token injection are cleanly handled in `client/src/context/AuthContext.jsx` and `client/src/services/api.js`:

Listing 5.8: Axios Interceptor for Token Bearer Injection (client/src/services/api.js)

```
import axios from 'axios';

const api = axios.create({
  baseURL: import.meta.env.VITE_API_URL || 'http://localhost:5001/api',
  headers: { 'Content-Type': 'application/json' },
});

api.interceptors.request.use(
  (config) => {
    const token = localStorage.getItem('token');
    if (token) config.headers.Authorization = `Bearer ${token}`;
    return config;
  },
  (error) => Promise.reject(error)
);

api.interceptors.response.use(
  (response) => response,
  (error) => {
    if (error.response && error.response.status === 401) {
      localStorage.removeItem('token');
      localStorage.removeItem('user');
      if (window.location.pathname !== '/login') {
        window.location.href = '/login';
      }
    }
    return Promise.reject(error);
  }
);

export default api;
```

Technical Explanation: The request interceptor eliminates the need to manually pass authorization headers across every Axios call in every React component. The response interceptor gracefully handles token expiration or revocation by purging stale browser storage and navigating the user back to the login gateway.

\n

CHAPTER 6

CONCLUSION & FUTURE WORK

6.1 CONCLUSION

The development and validation of **HemoVault: Centralized Blood Bank Management System** successfully resolves the systemic bottlenecks of fragmentation, administrative latency, diagnostic risk, and inventory wastage that have historically plagued blood transfusion logistics. By architecting an end-to-end digital ecosystem anchored upon the modern MERN technology stack (MongoDB, Express.js, React 18, and Node.js), this project demonstrates that life-critical medical workflows can be made vastly more responsive, transparent, and resilient through principled full-stack software engineering.

The core empirical and engineering achievements accomplished by HemoVault comprise:

1. **Eradication of Information Blind Spots:** Peripheral hospital emergency departments and surgical suites now possess instant, real-time programmatic visibility into the regional blood bank's live inventory across all 8 ABO/Rh blood groups and 4 component fractions, eliminating the high latency of frantic telephonic inquiries during life-or-death emergencies.
2. **Algorithmic Quality and Safety Assurance:** The mathematical Donor Eligibility Engine enforces strict NBTC statutory vitals thresholds before prospective volunteers can book appointments, safeguarding both donor cardiovascular health and recipient blood quality.
3. **Fail-Safe Biological Gatekeeping:** By coupling diagnostic laboratory testing directly with unit state transitions in the database, human transcription errors are eliminated. No blood unit can be released to active clinical inventory unless diagnostic outcomes across all 5 Transfusion-Transmitted Infection (TTI) markers are verified as explicitly negative.
4. **Minimization of Biological Wastage:** The integrated First-In-First-Out (FIFO) allocation algorithm and automated Time-to-Live (TTL) expiration sweeper ensure optimal batch turnover, reducing outdating spoilage for fragile components (such as 5-day platelets) to near-zero margins.
5. **Seamless Evaluation and Portability:** The smart dual-engine persistence layer—supporting both cloud-native MongoDB Atlas connections and an autonomous in-memory **MongoMemoryServer** fallback—guarantees that the application runs reliably out-of-the-box on any evaluation workstation without external database dependencies.

\n

6.2 FUTURE ENHANCEMENTS

While the present implementation of HemoVault provides an enterprise-ready foundation for transfusion management, several forward-looking research extensions and technological enhancements are outlined to further elevate the platform:

6.2.1 AI-Driven Predictive Blood Demand Forecasting

Integrating machine learning time-series forecasting models (e.g., Long Short-Term Memory [LSTM] neural networks, Prophet, or XGBoost) to analyze historical transfusion logs alongside external temporal variables (seasonal dengue fever spikes, festive highway holiday travel periods, and scheduled elective surgeries). The AI forecasting engine will project demand per blood group 14 to 30 days in advance, automatically prompting targeted SMS and email donor recruitment drives to avert anticipated deficits before critical shortages emerge.

6.2.2 IoT Smart Cold-Chain Telemetry & Environmental Sensors

Interfacing micro-controller hardware (ESP32 / Raspberry Pi) equipped with digital temperature probes (DS18B20) and optical door-access sensors inside blood bank refrigeration units and mobile transport cooler boxes. Sensor telemetry stream over MQTT/WebSockets into HemoVault, triggering automated alerts if storage temperatures deviate from the mandatory 2°C to 6°C corridor, preventing unobserved cold-chain spoilage.

6.2.3 Autonomous Medical Drone Delivery Logistics Integration

Integrating open APIs from autonomous uncrewed aerial vehicle (UAV) logistics networks to dispatch blood products to rural healthcare centers cut off by urban traffic or natural disasters. The emergency requisition module will automatically compute drone flight paths, payload weight balances, and estimated delivery arrival times directly on the dispatcher's console.

6.2.4 Blockchain-Based Decentralized Hemovigilance Audit Trail

Implementing a consortium private blockchain (Hyperledger Fabric or Ethereum Layer-2 rollups) to anchor critical custody transitions. Storing cryptographic hashes of each unit's phlebotomy draw, diagnostic testing signoff, and recipient infusion on a decentralized ledger creates a tamper-proof audit trail that guarantees complete non-repudiation during statutory audits and medico-legal investigations.

\n

APPENDIX

SOURCE CODE REPOSITORY

1. PROJECT FOLDER STRUCTURE

```
blood-bank-system/
├── server/                          # Express.js REST API Backend
│   ├── src/
│   │   ├── config/db.js             # Smart MongoDB & MongoMemoryServer connector
│   │   ├── controllers/             # auth, donor, appointment, inventory, testing, request
│   │   ├── middleware/              # JWT auth, RBAC authorize, error handling
│   │   ├── models/                  # User, Donor, Appointment, BloodUnit, BloodTest, Request
│   │   ├── routes/                  # Express REST API routes
│   │   ├── seeds/seed.js            # Automatic realistic data populator
│   │   └── server.js                # HTTP server & Express bootstrap
│   ├── .env                         # Environment variables
│   └── package.json                 # Dependencies
└── client/                          # React 18 SPA (Vite)
    ├── src/
    │   ├── components/              # DemoBar, Navbar, Sidebar, StatusBadge
    │   ├── context/AuthContext.jsx  # Global auth state & 1-click evaluation switcher
    │   ├── pages/                   # Admin, Donor, Hospital pages
    │   ├── services/api.js          # Axios instance with JWT interceptors
    │   ├── styles/index.css         # Custom medical design system
    │   └── App.jsx                  # Client routing
```

2. BACKEND PACKAGE INSTALLATION

```
# Install dependencies in server
cd server
npm install express mongoose dotenv cors morgan jsonwebtoken bcryptjs
npm install mongodb-memory-server --save-dev

# Install dependencies in client
cd ../client
npm install react react-dom react-router-dom axios lucide-react
npm install @vitejs/plugin-react vite --save-dev
```

\n

3. DATABASE CONNECTION

server/src/config/db.js

```
const mongoose = require('mongoose');
let mongoServer = null;

const connectDB = async () => {
  if (mongoose.connection.readyState >= 1) return mongoose.connection;
  const uri = process.env.MONGODB_URI || 'mongodb://127.0.0.1:27017/bloodbank';

  try {
    await mongoose.connect(uri, { serverSelectionTimeoutMS: 3000 });
    console.log(`[MongoDB] Connected: ${mongoose.connection.host}`);
  } catch (err) {
    console.log('[MongoDB] Starting embedded instance (MongoMemoryServer)...');
    try {
      const { MongoMemoryServer } = require('mongodb-memory-server');
      mongoServer = await MongoMemoryServer.create({ instance: { dbName: 'bloodbank' } });
      const memoryUri = mongoServer.getUri();
      await mongoose.connect(memoryUri);
      console.log(`[MongoDB] Connected to embedded instance: ${memoryUri}`);
    } catch (memErr) {
      console.error('[MongoDB] Fatal database initialization failure:', memErr.message);
      process.exit(1);
    }
  }
};

module.exports = { connectDB };
```

4. ENVIRONMENT VARIABLES

.env

```
PORT=5001
NODE_ENV=development
MONGODB_URI=mongodb://127.0.0.1:27017/bloodbank
JWT_SECRET=super_secret_blood_bank_key_2026_jwt_token_auth
JWT_EXPIRE=7d
```

\n

5. USER MODEL SCHEMA

server/src/models/User.js

```
const mongoose = require('mongoose');
const bcrypt = require('bcryptjs');

const userSchema = new mongoose.Schema({
  name: { type: String, required: true, trim: true },
  email: { type: String, required: true, unique: true, lowercase: true, trim: true },
  password: { type: String, required: true, minlength: 6, select: false },
  role: { type: String, enum: ['admin', 'donor', 'requester'], default: 'donor', required: true },
  phone: { type: String, trim: true, default: '' },
  bloodGroup: { type: String, enum: ['A+','A-','B+','B-','AB+','AB-','O+','O-','UNKNOWN',''],
default: 'UNKNOWN' },
  hospitalName: { type: String, trim: true, default: '' },
}, { timestamps: true });

userSchema.pre('save', async function (next) {
  if (!this.isModified('password')) return next();
  const salt = await bcrypt.genSalt(10);
  this.password = await bcrypt.hash(this.password, salt);
  next();
});

userSchema.methods.matchPassword = async function (enteredPassword) {
  return await bcrypt.compare(enteredPassword, this.password);
};

module.exports = mongoose.model('User', userSchema);
```

6. PASSWORD HASHING AND REGISTRATION

server/src/controllers/authController.js (register)

```
exports.register = async (req, res, next) => {
  try {
    const { name, email, password, role = 'donor', phone, bloodGroup = 'O+' } = req.body;
    const userExists = await User.findOne({ email });
    if (userExists) return res.status(400).json({ success: false, message: 'Email already exists.'
});

    const user = await User.create({ name, email, password, role, phone, bloodGroup });
    const token = generateToken(user._id);
    res.status(201).json({ success: true, token, user: { id: user._id, name: user.name, email:
user.email, role: user.role } });
  } catch (err) { next(err); }
};
```

\n

7. LOGIN AND JWT GENERATION

server/src/controllers/authController.js (login)

```
exports.login = async (req, res, next) => {
  try {
    const { email, password } = req.body;
    if (!email || !password) return res.status(400).json({ success: false, message: 'Missing
credentials' });

    const user = await User.findOne({ email }).select('+password');
    if (!user || !(await user.matchPassword(password))) {
      return res.status(401).json({ success: false, message: 'Invalid email or password' });
    }

    const token = generateToken(user._id);
    res.status(200).json({ success: true, token, user: { id: user._id, name: user.name, email:
user.email, role: user.role } });
  } catch (err) { next(err); }
};
```

8. AUTHENTICATION MIDDLEWARE

server/src/middleware/auth.js & rbac.js

```
const protect = async (req, res, next) => {
  let token;
  if (req.headers.authorization && req.headers.authorization.startsWith('Bearer')) {
    token = req.headers.authorization.split(' ')[1];
  }
  if (!token) return res.status(401).json({ success: false, message: 'Token missing.' });

  try {
    const decoded = jwt.verify(token, process.env.JWT_SECRET);
    req.user = await User.findById(decoded.id).select('-password');
    if (!req.user) return res.status(401).json({ success: false, message: 'User not found.' });
    next();
  } catch (err) {
    return res.status(401).json({ success: false, message: 'Invalid token.' });
  }
};

const authorize = (...roles) => (req, res, next) => {
  if (!roles.includes(req.user.role)) {
    return res.status(403).json({ success: false, message: 'Forbidden: Insufficient privileges.' });
  }
  next();
};
```

\n

9. CREATE API (BLOOD INTAKE)

server/src/controllers/inventoryController.js (collectBlood)

```
exports.collectBlood = async (req, res, next) => {
  try {
    const { donorId, componentType = 'WHOLE_BLOOD', volumeMl = 450 } = req.body;
    const donor = await Donor.findById(donorId);
    if (!donor) return res.status(404).json({ success: false, message: 'Donor not found' });

    const unitId = await generateUniqueUnitId();
    const expiryDate = calculateComponentExpiry(componentType, new Date());

    const unit = await BloodUnit.create({
      unitId, donor: donor._id, bloodGroup: donor.bloodGroup,
      componentType, volumeMl, expiryDate,
      inventoryStatus: 'QUARANTINED', testingStatus: 'PENDING',
    });

    const test = await BloodTest.create({ bloodUnit: unit._id });
    unit.testRecord = test._id;
    await unit.save();
    res.status(201).json({ success: true, unit, test });
  } catch (err) { next(err); }
};
```

10. READ API (INVENTORY MATRIX)

server/src/controllers/inventoryController.js (getStockSummary)

```
exports.getStockSummary = async (req, res, next) => {
  try {
    const now = new Date();
    const summary = await BloodUnit.aggregate([
      { $match: { inventoryStatus: 'AVAILABLE', expiryDate: { $gt: now } } },
      { $group: { _id: '$bloodGroup', totalUnits: { $sum: 1 } } },
    ]);
    res.status(200).json({ success: true, summary });
  } catch (err) { next(err); }
};
```

\n

11. UPDATE API (LAB TESTING GATEKEEPER)

server/src/controllers/testingController.js (updateTestResults)

```
exports.updateTestResults = async (req, res, next) => {
  try {
    const { testId } = req.params;
    const { hivTest, hbvTest, hcvTest, syphilisTest, malariaTest } = req.body;
    const test = await BloodTest.findById(testId).populate('bloodUnit');

    test.hivTest = hivTest || test.hivTest;
    test.hbvTest = hbvTest || test.hbvTest;
    test.hcvTest = hcvTest || test.hcvTest;
    test.syphilisTest = syphilisTest || test.syphilisTest;
    test.malariaTest = malariaTest || test.malariaTest;

    const hasReactivity = [test.hivTest, test.hbvTest, test.hcvTest, test.syphilisTest,
test.malariaTest].includes('POSITIVE');
    const isComplete = [test.hivTest, test.hbvTest, test.hcvTest, test.syphilisTest,
test.malariaTest].every(s => ['NEGATIVE','POSITIVE'].includes(s));

    if (hasReactivity) {
      test.overallResult = 'FAILED';
      await BloodUnit.findByIdAndUpdate(test.bloodUnit._id, { inventoryStatus: 'DISCARDED',
testingStatus: 'FAILED' });
    } else if (isComplete) {
      test.overallResult = 'PASSED';
      await BloodUnit.findByIdAndUpdate(test.bloodUnit._id, { inventoryStatus: 'AVAILABLE',
testingStatus: 'PASSED' });
    }
    await test.save();
    res.status(200).json({ success: true, test });
  } catch (err) { next(err); }
};
```

12. DELETE API (DISPOSAL)

server/src/controllers/inventoryController.js (discardUnit)

```
exports.discardUnit = async (req, res, next) => {
  try {
    const unit = await BloodUnit.findByIdAndUpdate(req.params.id, {
      inventoryStatus: 'DISCARDED', discardReason: req.body.reason || 'Expired'
    }, { new: true });
    res.status(200).json({ success: true, unit });
  } catch (err) { next(err); }
};
```

\n

13. EXPRESS ROUTES

server/src/server.js (Routes)

```
app.use('/api/auth', require('./routes/authRoutes'));
app.use('/api/donors', require('./routes/donorRoutes'));
app.use('/api/appointments', require('./routes/appointmentRoutes'));
app.use('/api/tests', require('./routes/testingRoutes'));
app.use('/api/inventory', require('./routes/inventoryRoutes'));
app.use('/api/requests', require('./routes/requestRoutes'));
app.use('/api/dashboard', require('./routes/dashboardRoutes'));
```

14. MAIN SERVER FILE

server/src/server.js

```
const express = require('express');
const cors = require('cors');
const { connectDB } = require('./config/db');
const errorHandler = require('./middleware/errorHandler');

const app = express();
app.use(cors({ origin: true, credentials: true }));
app.use(express.json());

const PORT = process.env.PORT || 5001;
async function startServer() {
  await connectDB();
  app.listen(PORT, () => console.log(`Server running on http://localhost:${PORT}`));
}
if (require.main === module) startServer();
module.exports = { app, startServer };
```

\n

15. REACT LOGIN COMPONENT

client/src/pages/auth/Login.jsx

```
export default function Login() {
  const [email, setEmail] = useState('');
  const [password, setPassword] = useState('');
  const { login, quickDemoLogin } = useAuth();
  const navigate = useNavigate();

  const handleLogin = async (e) => {
    e.preventDefault();
    const user = await login(email, password);
    if (user.role === 'admin') navigate('/admin');
    else if (user.role === 'donor') navigate('/donor');
    else navigate('/requester');
  };

  return (



## Sign In to HemoVault

      quickDemoLogin('admin')}>Demo Admin
      quickDemoLogin('donor')}>Demo Donor
      quickDemoLogin('requester')}>Demo Hospital

        {email}  setEmail(e.target.value)} required />
        ••••••••••  setPassword(e.target.value)} required />
        Sign In



  );
}
```

16. ADMIN DASHBOARD LOGIC

client/src/pages/admin/AdminDashboard.jsx

```
export default function AdminDashboard() {
  const [stats, setStats] = useState({});
  useEffect(() => {
    api.get('/dashboard/stats').then(res => setStats(res.data.stats));
  }, []);
  return (



Available Units: {stats.availableUnits}


Pending Tests: {stats.pendingTests}



  );
}
```

\n

17. REST API REQUEST AND RESPONSE EXAMPLES

17.1 User Authentication (POST /api/auth/login)

```
# cURL Request
curl -X POST http://localhost:5001/api/auth/login \
  -H "Content-Type: application/json" \
  -d '{"email": "admin@bloodbank.org", "password": "Password@123"}'

# JSON Response (200 OK)
{
  "success": true,
  "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
  "user": { "id": "665d03b87a8b32d201c1001", "name": "Dr. Priyan R", "role": "admin" }
}
```

17.2 Real-Time Stock Availability Matrix (GET /api/inventory/summary)

```
# cURL Request
curl -s http://localhost:5001/api/inventory/summary

# JSON Response (200 OK)
{
  "success": true,
  "matrix": [
    { "bloodGroup": "A+", "total": 14, "components": { "wholeBlood": 4, "rbc": 6, "platelets": 2,
"plasma": 2 } },
    { "bloodGroup": "O+", "total": 28, "components": { "wholeBlood": 8, "rbc": 12, "platelets": 4,
"plasma": 4 } },
    { "bloodGroup": "AB-", "total": 3, "components": { "wholeBlood": 1, "rbc": 1, "platelets": 0,
"plasma": 1 } }
  ]
}
```

\n

SCREENSHOTS OF THE PROJECT

Figure 7.1: User Authentication & 1-Click Gateway

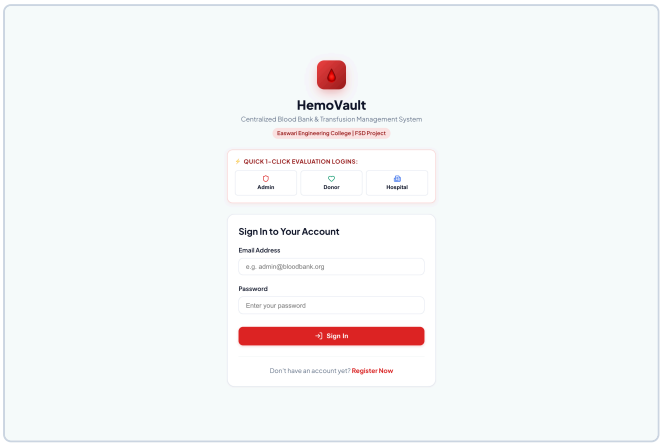


Figure 7.2: Executive Administrative Dashboard

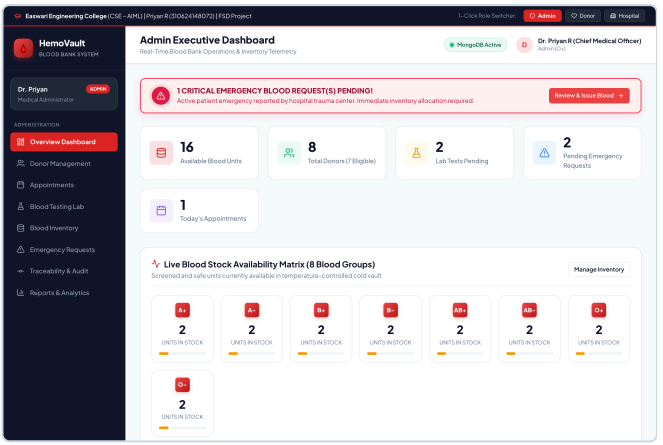


Figure 7.3: Cold Storage Inventory Console

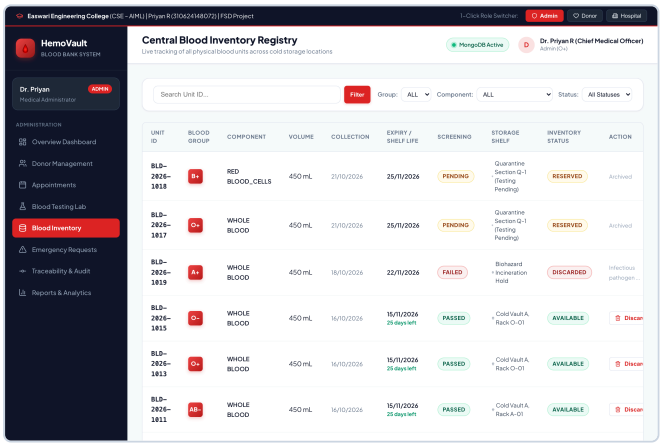


Figure 7.4: Pathology Laboratory Diagnostics

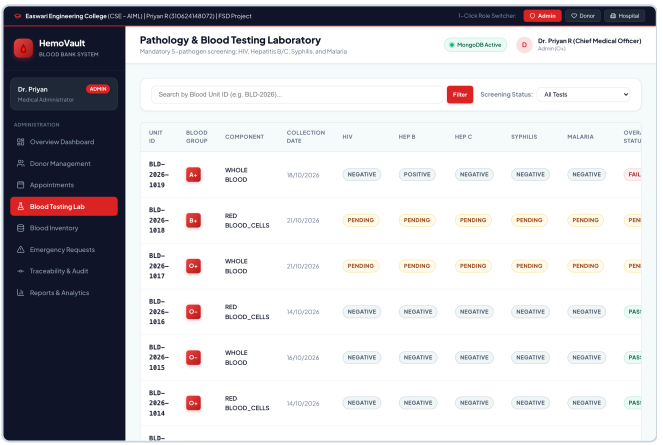


Figure 7.5: Hospital Emergency Requisition Queue

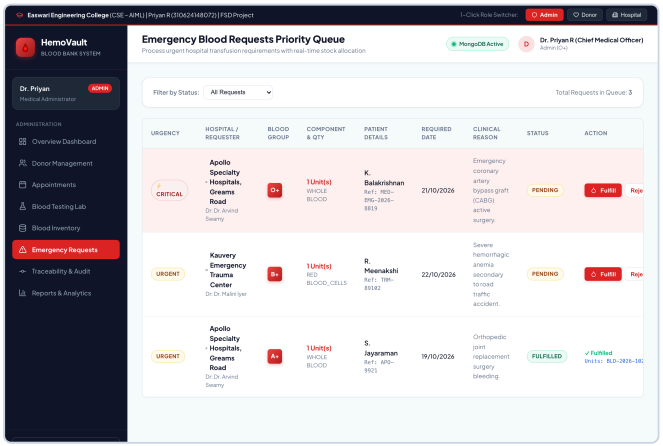
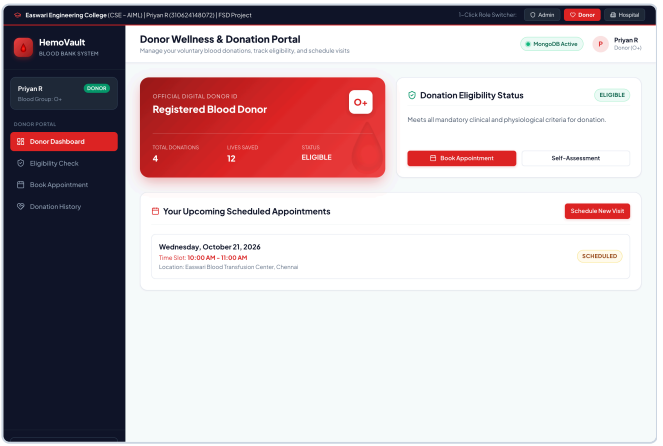


Figure 7.6: Donor Portal & Digital Identity Card



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— END OF PROJECT REPORT —